Assignment 1: Small Transformer Language Models

Lior Alafi 032543472, Tal Somech 315473033

23 April 2026

The Data

The English dataset comprises a vocabulary of 66 tokens and does not include unusual or non-standard characters. It consists of approximately nine works attributed to Shakespeare.

The Hebrew dataset contains a vocabulary of 89 tokens, including approximately 70 non-Hebrew characters, primarily from the Cyrillic alphabet. Furthermore, the dataset was not fully cleaned and includes residual content originating from the Ben Yehuda Project website (e.g., navigation menus and boilerplate text).

We conducted experiments both with a cleaned version of the dataset—removing these artifacts—and with the raw, unprocessed data. While retaining such artifacts introduces bias, it also provides valuable insight into the impact of noise on model behavior.

residuals in Hebrew dataset

לשאת רחל פרויקט בן-יהודה.
דף זה הוא חלק מפרויקט בן-יהודה.
לתוכן העניינים לדף הראשי של פרויקט בן-יהודה.
דף זה הוא חלק מפרויקט בן-יהודה, שמתקיים הודות לסיוע של מתנדבים.

The Hebrew dataset comprises 495 poems by Rachel, along with works by Bialik. The tokenizer provided for this task is character-level, processing the text one character at a time.

Building the Model

We identified a limitation in the provided implementation instructions. Specifically, the function `TransformerLM::init_weights` was implemented by iterating over `named_parameters()`. However, in PyTorch, the elements returned by `named_parameters()` are instances of `nn.Parameter` and do not retain information about the originating module type (e.g., `nn.Linear`). Consequently, this approach does not support proper layer-specific initialization.

To address this issue, we modified the implementation to iterate over the model's modules instead, thereby enabling correct initialization of layer types such as `nn.Linear`.

In addition, we introduced several enhancements to the model, including support for extracting attention maps, integration of dropout layers, and improved GPU compatibility. We also adopted a pre-normalization (pre-norm) architecture, as it has been shown to improve training stability, particularly in deeper transformer models, and is widely used in modern implementations.

1 Chosen Parameters

We split the dataset into 90% training data and 10% validation data. Both the Hebrew and English models underwent the same training and hyperparameter optimization process. To identify optimal configurations, we employed both Optuna-based optimization and random search. In addition, we used a learning rate scheduler (`ReduceLROnPlateau`) and implemented early stopping.

Experiments were conducted on two different hardware setups: a local machine equipped with an NVIDIA RTX 4080 (laptop) and Google Colab using a T4 GPU. The random search experiments were primarily executed on the RTX 4080, while Optuna-based optimization was performed on Colab.

Although Optuna yielded superior results, we present both approaches along with our observations and conclusions regarding the hyperparameter search process.

Hyperparameter	**Search Space**	**Sampling Strategy**
Sequence Length (seq_len)	$\{128, 256\}$	Uniform random choice
Batch Size ($batch\_size$)	$\{32, 64, 128\}$	Uniform random choice
Number of Layers (n_layers)	$\{4, 6, 8\}$	Uniform random choice
Number of Heads (n_heads)	$\{4, 6, 8\}$	Uniform random choice
Embedding Size ($embed\_size$)	$\{64, 128, 192, 256, 384\}$	Uniform random choice
MLP Hidden Size	$4 \cdot d_{model}$	Deterministic
Learning Rate	$[5 \cdot 10^{-7}, 10^{-4}]$	Log-uniform sampling
Dropout	$\{0, 0.05, 0.1, 0.2\}$ (per layer)	Independent random per component
Constraint	$embed\_size \bmod n\_heads = 0$	Enforced post-sampling

Table 1: Hyperparameter search space and sampling strategy

Shakespeare (English)

We initially trained the model for an extended period without modifying the hyperparameters, in order to assess its baseline performance. The model reached a validation loss below 2.0 after approximately 200 training steps and achieved a validation loss of 1.5 within 3,000 batches. Beyond this point, signs of overfitting began to emerge. Based on this observation, we determined that 7,000 batches would be sufficient for hyperparameter search.

Using Optuna for hyperparameter optimization, we obtained a validation loss of 1.3972. However, when evaluating the hyperparameters found for the Hebrew dataset on the English dataset, we observed improved performance, achieving a best validation loss of 1.3795 after 5,300 batches (see Figure 1).The training corpus comprises **1,115,394 characters**, yielding approximately **8,700 training sequences** of length 128 (90% split). The corresponding model has approximately 5.86M parameters. We also observed that a dropout rate of 0.2 contributed to improved generalization and lower validation loss.

The final configuration used is as follows:

1. Sequence length: 128
2. Batch size: 32
3. Number of layers: 8
4. Number of heads: 4
5. Embedding size: 256
6. MLP hidden size: 1024
7. Learning rate: 5×10^{-4}

8. Dropout rate: 0.2
9. Residual connections enabled

This result also highlights the limitations of automated hyperparameter search, as manually transferred configurations outperformed those obtained via Optuna in this case.

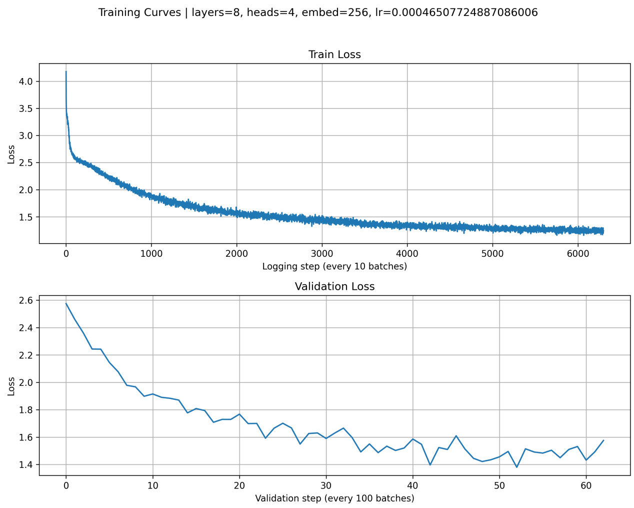


Figure 1: english loss

Hyperparameters

As described in Table 1, we performed approximately 30 trials using random search (both with and without Optuna). The figures presented here represent a subset of the experiments and are not exhaustive.

We observe that, in the absence of residual connections, the model fails to learn, as shown in Figure 3, despite the fact that the same hyperparameter configurations with residual connections exhibit effective learning. Residual connections are essential, as they provide a direct pathway for both activations and gradients across layers. Instead of requiring each Transformer block to fully reconstruct the representation, each block learns a residual correction:

$$x_{l+1} = x_l + F(x_l).$$

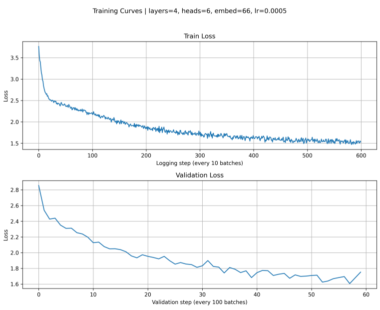
This formulation stabilizes optimization, particularly in deeper architectures. Without residual connections, information must propagate through a long sequence of nonlinear transformations, which leads to poor gradient flow and ineffective training. This explains the near-constant training and validation loss observed in the no-residual setting.

Another critical hyperparameter is the embedding dimension. Smaller values (e.g., 64–66) produced stable training but exhibited signs of underfitting (see Table 1**A**). In contrast, larger values (e.g., above 256) consistently led to overfitting, regardless of the number of attention heads (see Table 1**E,F**).

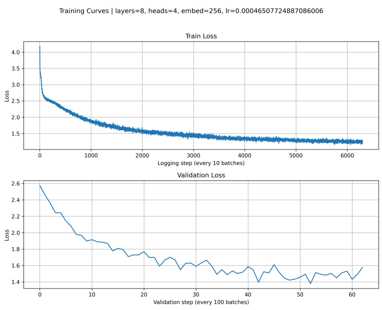
We also observed that increasing the number of attention heads does not necessarily improve performance. A large number of heads may result in relatively low-dimensional ("weak") individual heads, limiting their representational capacity. Even when selecting configurations with fewer but larger heads, no significant improvement was observed.

In summary, the embedding dimension exhibits a clear "sweet spot" in the range of 128–256, where models demonstrate stable training dynamics and balanced generalization performance, both with and without dropout.

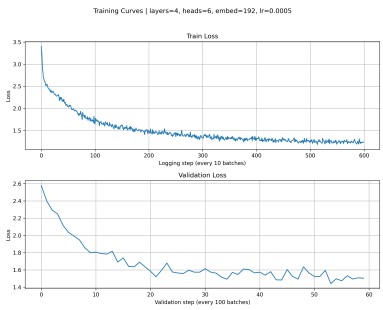
This behavior is consistent with the trade-off between model capacity and generalization: insufficient dimensionality limits representational power, while excessive dimensionality increases the risk of overfitting given the dataset size.



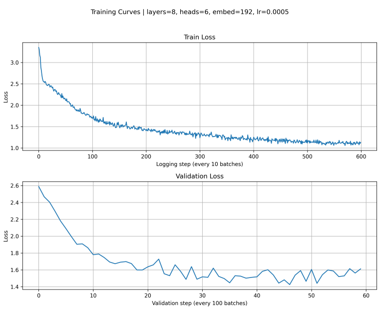
(a) A



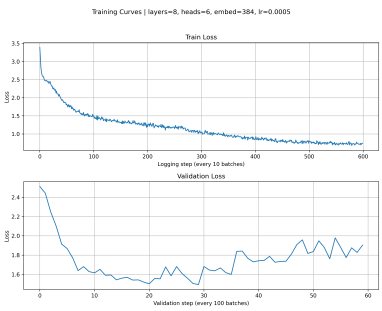
(b) B



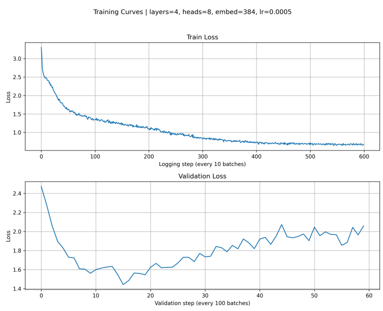
(c) C



(d) D



(e) E



(f) F

Figure 2: Hyperparameter search

From Figure 2, we observe that configurations with 4–8 layers exhibit stable and effective learning dynamics. In contrast, the number of attention heads is best constrained to the range of 4–6, as larger values did not provide consistent improvements and in some cases degraded performance.

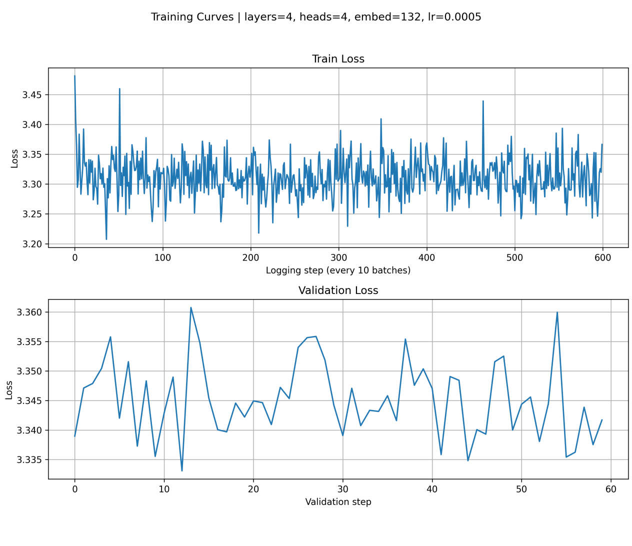
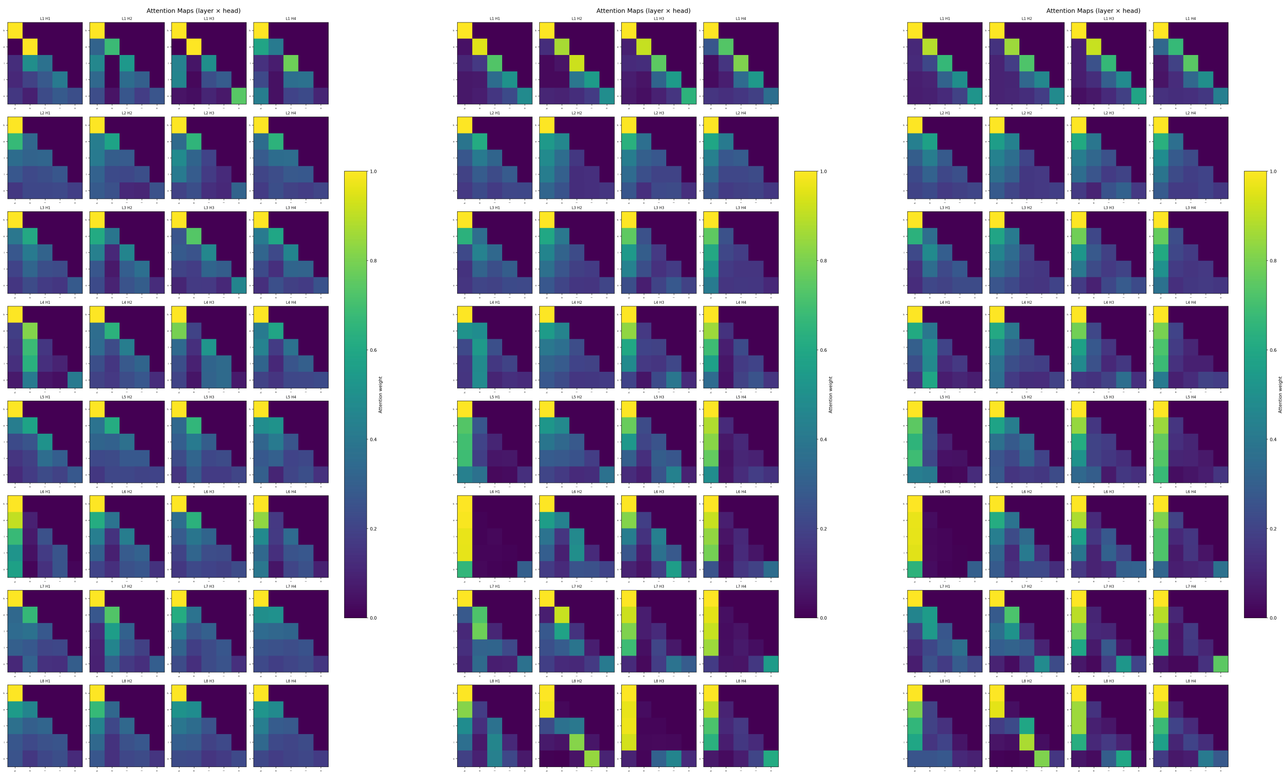


Figure 3: No residual effect

Furthermore, we examined how attention maps evolve over the course of training, in order to better understand the model's internal representation dynamics:



(a) batch 200 (b) batch 3200 (c) batch 6300

Figure 4: Attention maps while training

During the early stages of training, as shown in Figure 4**a**, many attention heads tend to focus disproportionately on the first token. This behavior likely reflects an initial bias before meaningful patterns are learned.

In contrast, at a later stage of training (Figure 4**b**), the attention patterns become more structured. In particular, the lower layers exhibit both backward-looking attention (manifested as vertical stripe patterns) and local attention (diagonal patterns), indicating that the model begins to capture short-range dependencies and contextual relationships.

An additional aspect of interest is the model's ability to generate coherent text during training. However, evaluating generation quality within this setup is non-trivial. In particular, it is difficult to quantitatively assess whether the generated sequences are meaningful. Ideally, this could be addressed by incorporating an auxiliary evaluation mechanism, such as a teacher model or external scoring function, capable of estimating the semantic coherence or fluency of generated text.

This highlights a limitation of relying solely on loss-based metrics, as they do not necessarily capture the qualitative aspects of generation, such as coherence or linguistic plausibility.

Bialik & Rachel (Hebrew)

We initially trained the model without modifying hyperparameters to establish a baseline. The model reached validation loss below 2.2 after approximately 200 steps and achieved 1.97 within 3,000 batches, after which overfitting became apparent. Based on this, we limited the hyperparameter search to 6,000 batches.

The best validation loss obtained was **1.8724**, achieved after **1,500 batches**. This suggests that improvement seen up to 3,000 batches in earlier experiments was likely due to suboptimal initialization causing delayed overfitting. A dropout rate of 0.2 consistently improved generalization.

The Hebrew dataset presents additional challenges compared to English: richer morphology and more compact word structure increase token-distribution complexity. This is directly evidenced in our results: higher validation loss (1.87 vs. 1.38), a higher rate of hallucinated words in generated outputs, noisier training data, and less coherent samples across all temperatures. The training corpus comprises approximately **1,500,000 characters**, yielding approximately **10,500 training sequences** of length 128 (90% split).

The final configuration is identical to the English model **5.86M parameters**, 8 layers, 4 heads, embed=256, MLP=1024, lr=5×10^{-4}, dropout=0.2, seq_len=128, batch=32.

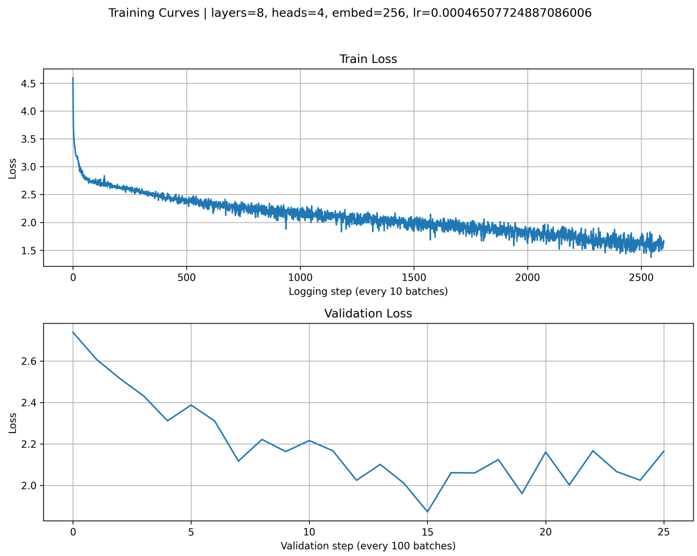
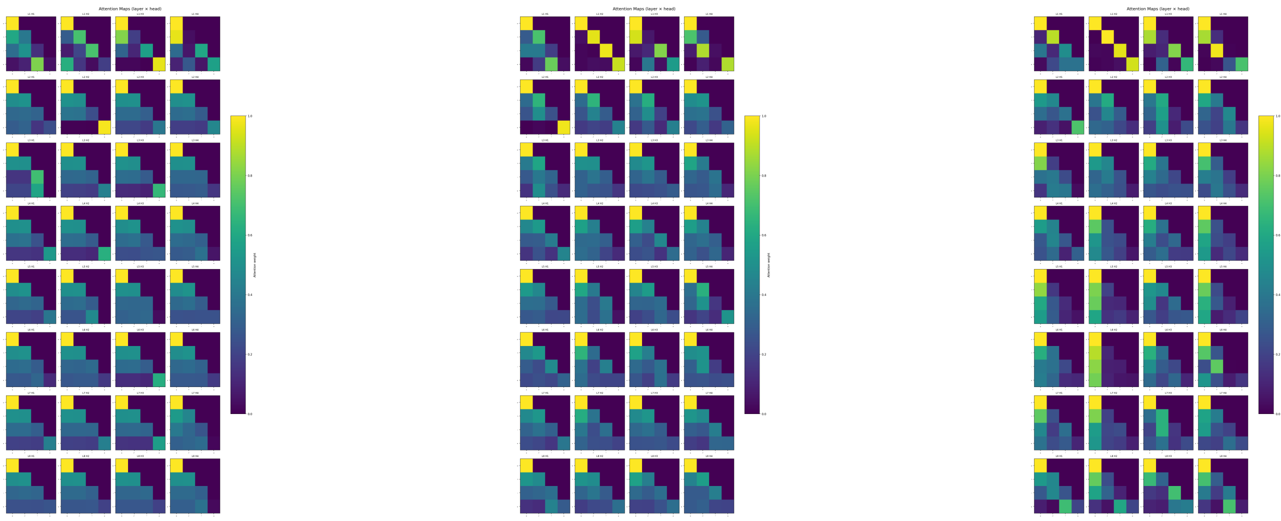


Figure 5: Training and validation loss curves for the final Hebrew model (same architecture as English). Best validation loss: 1.8724 at batch 1,500.

To analyze attention evolution, we evaluated the model using the Hebrew prefix שלום at intervals of 100 training batches.

In the early stages (Figure 6a), many attention heads exhibit similar behavior, indicating a lack of specialization. At a later stage (Figure 6b), patterns begin to differentiate: **Layer 1, Head 2** focuses on local context, while other heads retain more uniform behavior. By a more advanced stage (Figure 6c), clear specialization emerges—each head captures a distinct aspect of the input. Specifically, Layer 1 Head 2 attends to local character-level features, while deeper layers (e.g., Layer 8) tend to attend to the beginning of the word.

Comparison: Hebrew vs. English Compared to English, the Hebrew model shows slower and less distinct head specialization. This aligns with the richer morphology and higher character-level variability of Hebrew, which makes learning consistent local patterns more difficult. The higher validation loss and greater incidence of hallucinated tokens in Hebrew generation further support this interpretation.



(a) batch 100 (b) batch 700 (c) batch 1500

Figure 6: Attention maps while training

Final Results Summary

Property	English (Shakespeare)	Hebrew (Bialik & Rachel)
Best validation loss	1.3795	1.8724
Batches at best val. loss	5,300	1,500
Parameters	5.86M	5.86M
Sequence length	128	128
Batch size	32	32
Layers	8	8
Heads	4	4
Embedding dim	256	256
MLP hidden dim	1024	1024
Dropout	0.2	0.2
Learning rate	5×10^{-4}	5×10^{-4}
Vocab size	66	89
Corpus size (characters)	1,115,394	∼1,500,000
Approx. training sequences	∼8,700	∼10,500

Table 2: Final results summary for both language models.

2 English model Generation , Inference and Analysis

Generation Analysis

English was easier to train, but it is also clear that character-level tokens adapt better when it comes to English.

In order to better understand and assess the generation process, we took a bank of prefixes (e.g., "Hello", "Where art thou") and ran our best model with temperatures ranging from 0.1 to 1.0 (in increments of 0.2), using top-k = 5 as suggested. We generated samples using `better_sample_continuation` (complex) and `sample_continuation` (simple).

Morphology(word structure)
lexical(word meaning)
Syntactic(sentence structure)
Semanti(sentence meaning)
prefix connection
Prefix: `where art thou`

No Temperature (default)

```
come-ave it
Is it was an accusation: or rein I'll speak
The fair prodigal of my bame your queen,
Nor o'er but in thy deep; and Signior Him,
With tears the weigh.
Come you, fearing?

PAULINA:
Never well, you will be buttless.

DUKE VINCENTIO:
For nurse, he
```

Temperature = 0.10

```
art a bawd,
And the seal'd the sea of the seal of his head,
And therefore the seat of the base of the sea
And the seas of the seas of the seas.

CORIOLANUS:
The prince of the common of the world,
And there is the sea of the seal's son,
And there I was the
```

Temperature = 0.30

```
art and the seast,
That thou hast stand with thy brother's head;
And then the consented with the seal's sea,
And then being so such a shame and made the friends
And strike an earth in the contraction
That should be so much as a banished and heaven
That th
```

Temperature = 0.50

```
wilt to the contract,
And beg my heaven so better'd the state,
Than my father be so sharp thee to the contract
Thou art the war'st the way to his happy hand.
```

```
Lord Mayor:
The steel of the world to my heart.

PAULINA:
Nay, but I am sorry the consent and my
```

Temperature = 0.70

```
the power of the breather,
The princes of the wool'st death body was a cause
To be so themselves of the body the way,
That thou shalt be been mine to this feeling speak.

LUCIO:
In the sea are and mountain's honour of a will.

DUKE VINCENTIO:
I have neith
```

Temperature = 0.90

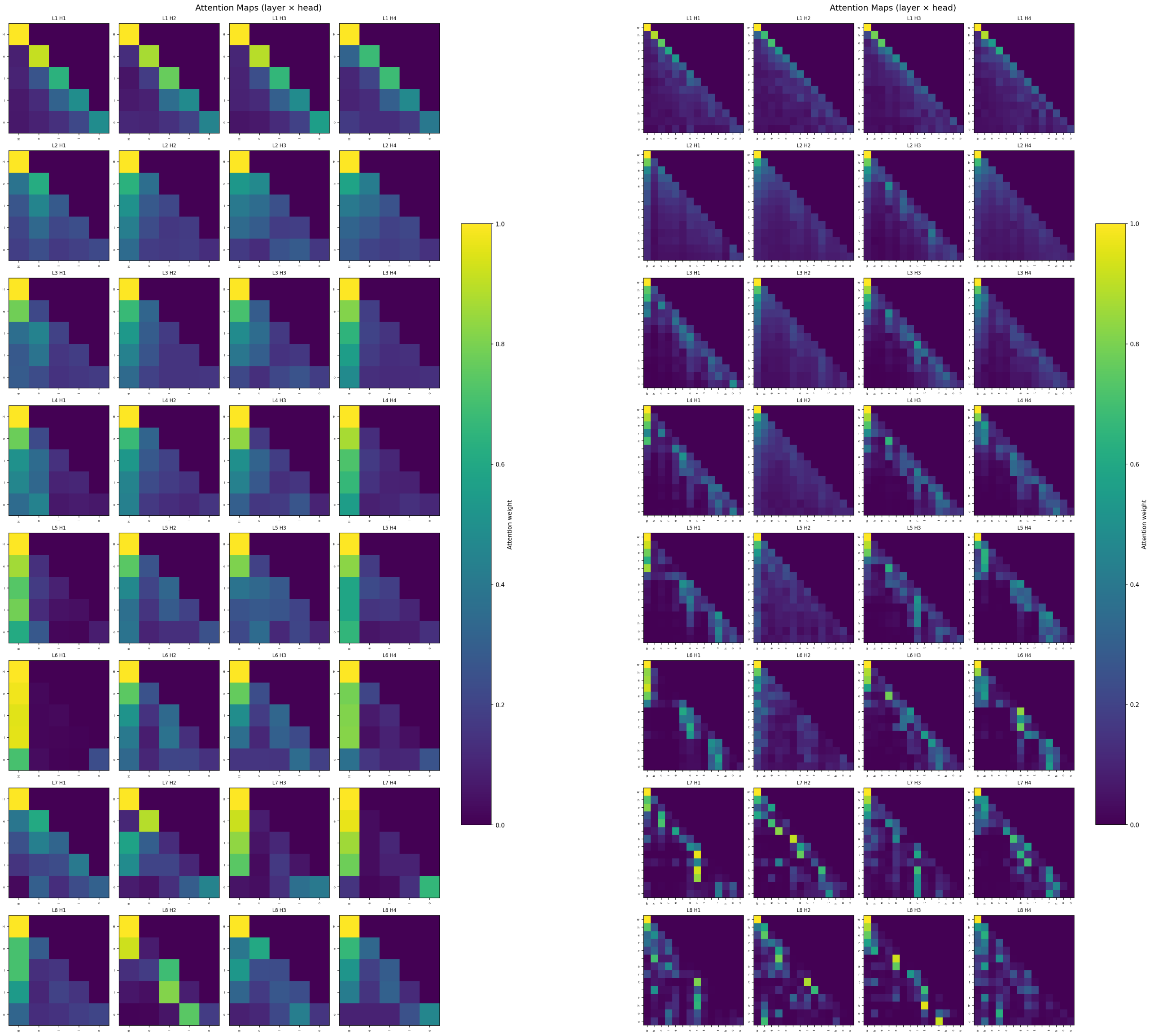
```
speak'st and shrewd.

GLOUCESTER:
I think thee, beseech you, as these wonder'd.

Lord:
His is that the charest wild with this change.

DUKE VINCENTIO:
This is that second more to hear the blood
With husband'st here to be seen: that will he shall be
to ban
```

Comparative Analysis by Temperature

Temperature	Characteristics	Main Issues
Default	Moderate diversity; partial Shakespearean tone; some dialogue structure	Hallucinated/corrupted words (e.g., "greech"), weak syntax, low semantic coherence
0.10	Very close to model distribution; highly deterministic; repetitive patterns (e.g., "the sea")	Severe repetition, low diversity, semantic stagnation
0.30	Slightly more diverse; retains some structure and tone	Still repetitive; grammatical errors; limited coherence
0.50	Balanced diversity; clearer sentence structure; improved readability	Inconsistent grammar; shallow meaning; partial incoherence
0.70	High diversity; more creative and varied phrasing	Increased syntactic errors; semantic drift; reduced stability
0.90	Very diverse; richer vocabulary; stronger Shakespearean flavor	Highly unstable; incoherent sentences; many grammatical errors

Summary: Lower temperatures produce outputs closer to the model's highest-probability predictions, resulting in more stable but repetitive text. Higher temperatures increase diversity and creativity, but introduce more grammatical errors and reduce semantic coherence.

Attention maps Analysis



(a) Prefix: “Hello”

(b) Prefix: “Where art thou”

Figure 7: Attention maps of the final English model for different input prefixes.

In **Hello** We observe strong attention to the first token, suggesting that some heads use it as a global reference point. Additionally, diagonal patterns indicate local dependencies, where each token attends primarily to nearby previous tokens (e.g. Layer 1 Head 2 attends mainly to the previous character).

while in **where art thou** We observe attention patterns linking ”where” to ”ar” in Layer 8H1, which may reflect statistical patterns learned from the training data (e.g., the frequent phrase ”where are”). This suggests that the model captures subword or character-level correlations rather than true semantic understanding.

Overall, the attention maps indicate that the model relies heavily on positional structure and statistical co-occurrence patterns in the data.

3 Hebrew Model Performance Comparison

We evaluated the Hebrew model using both simple and complex generation methods with the prefix שלום and a maximum length of 128. While the simple implementation does not use temperature, the complex model (top-k=5) benefits from temperature scaling.

At low temperature (0.10), the complex model produces repetitive and less coherent outputs, whereas the simple method yields slightly more stable text:

Simple Implementation:

בעל תרפשים .
פרדות לי לא אשר מעוה,
בצנרניך בקל פרחיץ הרחנים .
ובלי מתחידות .
והחלום יטוב דמעיר.
אכהות דורת לבי וחרחוב

Temperature 0.10

ובא היום המרחוק .
ובמלא השמש המרחב .
ובמלא היום המרחוק .
ובשל המרחוק המרחוק .

At moderate temperature (0.50), the complex model achieves improved fluency and reduced repetition:

Temperature 0.50

לא את את לבבי .
אל האחות השמתים השלום .
ואנכי ביתי הריחוקה והדוים השמתי על .
בשלח השל הריח הרוע לילה .
לעל כל כבוד המלאה

At higher temperature (0.90), the complex model produces more diverse and expressive outputs, albeit with increased noise:

Temperature 0.90

אחרתי אלי .
מרחב הרחמה .
ומבט לעבודה .
מעביבים .
ומלבי של השל .
ומי לך של אחות .
ומשור לא נפשי לא אחרת.
ואחת מאדים

Bialik & Rachel Style Analysis

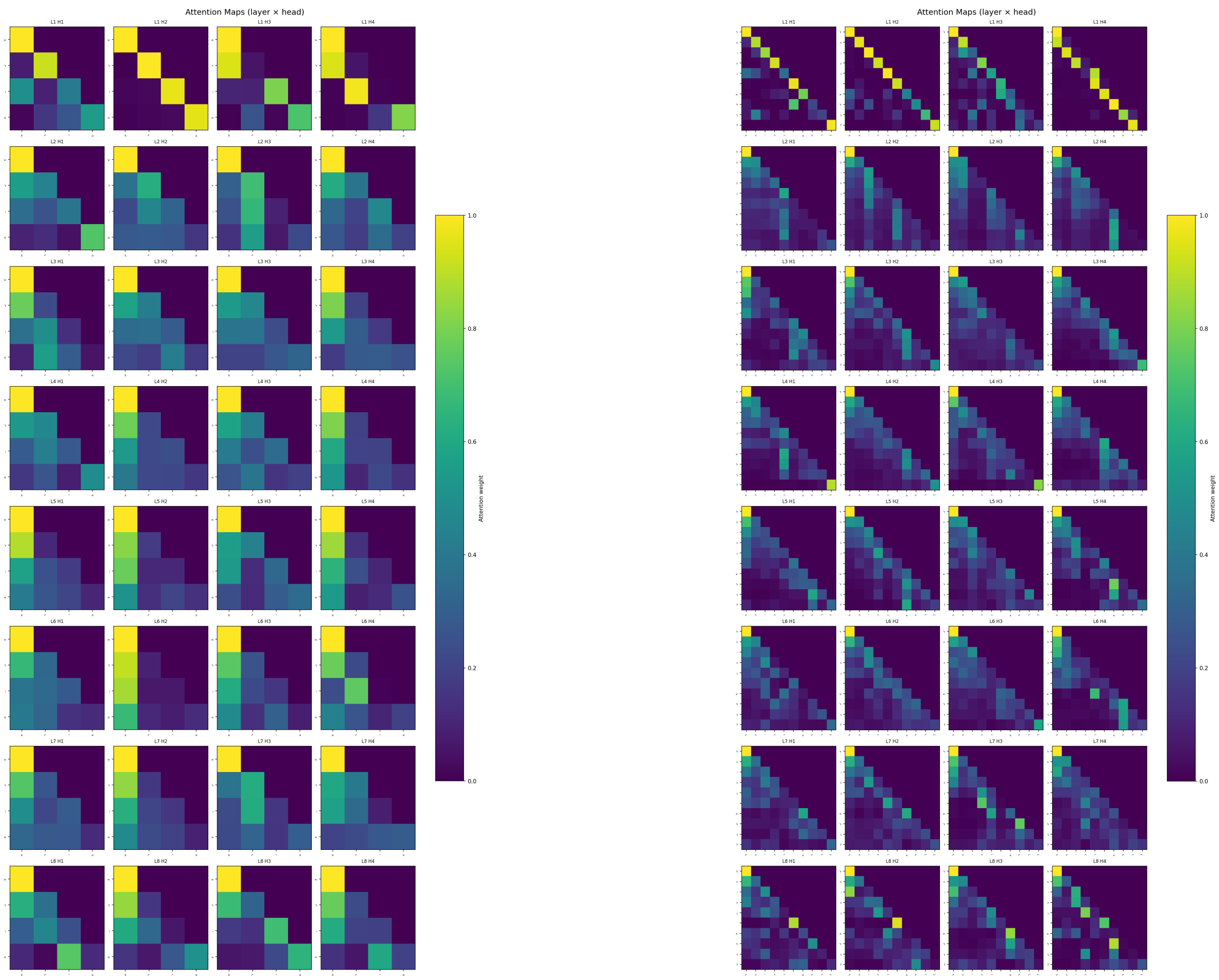
Both generation methods capture surface-level characteristics of Hebrew poetic language, such as the use of emotional vocabulary (לב, נפש, מרחב) and frequent prefix chaining. However, the generated text lacks deeper structural properties.

Lexically, the model mixes valid poetic terms with hallucinated words, likely due to character-level tokenization. Syntactically, the output consists of fragmented phrases rather than coherent sentences, lacking the hierarchical structure observed in authentic poetry. Semantically, there is no consistent narrative or emotional progression, resulting in static and unstructured meaning.

Overall, the model successfully mimics stylistic surface patterns but fails to capture the underlying grammatical and compositional rules characteristic of Bialik and Rachel.

3.1 Attention Maps Analysis

We analyzed attention maps for both single-word and longer inputs. Early layers tend to focus on local regions or initial tokens, while deeper layers distribute attention more broadly across the sequence. In longer inputs, certain heads (e.g. Layer 8 Head 1) attends strongly to spaces/punctuation, suggesting that the model partially relies on structural cues rather than purely semantic information.



(a) batch 100 (b) batch 700

Figure 8: Attention maps while generating

Limitations

Several limitations should be noted. First, evaluation of generated text is primarily qualitative, making it difficult to objectively assess coherence and linguistic quality. Second, the use of character-level tokenization increases the likelihood of generating invalid or hallucinated words, particularly in morphologically rich languages such as Hebrew. Third, the relatively small dataset size limits the model's ability to learn deeper syntactic and semantic structures, leading to surface-level imitation rather than true stylistic understanding. Finally, attention map analysis provides useful intuition but does not fully explain the model's internal decision-making process, and should therefore be interpreted with caution.

4 Conclusion and Summary

In this work, we implemented and analyzed a character-level Transformer language model for both English and Hebrew datasets. We explored the effects of various architectural and training choices, including residual connections, embedding size, number of layers, and dropout. Our experiments demonstrate that residual connections are critical for stable training, and that model capacity must be carefully balanced to avoid underfitting and overfitting.

Through hyperparameter optimization, we identified a stable configuration that generalizes well across both languages. Interestingly, parameters tuned on the Hebrew dataset achieved competitive, and in some cases superior, performance on English, suggesting a degree of robustness in the selected architecture.

We further investigated the model's internal behavior using attention maps, observing a clear progression from unstructured attention in early training to more specialized and diverse patterns in later stages. This supports the interpretation that Transformer models gradually learn hierarchical representations.

In terms of generation, we found that temperature plays a crucial role in balancing coherence and diversity. While low temperatures produce repetitive and rigid outputs, higher temperatures enable more expressive generation at the cost of increased noise. Despite capturing surface-level stylistic patterns, the model struggles to generate fully coherent and semantically consistent text, particularly in Hebrew.

Overall, our results highlight both the strengths and limitations of small-scale Transformer models. While they are capable of learning meaningful patterns and producing stylistically plausible text, their ability to capture deeper linguistic structure remains constrained by dataset size, tokenization choice, and model capacity.